

# **Capacity, Not Destiny:**

*A Comparative Neuroanatomical Hypothesis on Olfaction,  
Memory, and the Homo sapiens–Neanderthal Divergence*

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## Abstract

This paper is a conceptual and theoretical synthesis, not new anatomical or experimental data: it connects three separate existing literatures — comparative endocranial anatomy, olfactory neuroanatomy, and rodent neuroplasticity — into a single testable hypothesis, and states plainly where that connection is supported and where it is not. *Homo sapiens* and *Homo neanderthalensis* evolved brains of comparable overall size through independent evolutionary trajectories, yet three-dimensional reconstruction of the fossil endocranial base indicates a structural divergence: *Homo sapiens* show relatively larger olfactory-bulb fossae and a more expanded, forward-projecting temporal pole than Neanderthals (Bastir et al., 2011; Kochiyama et al., 2018) — a finding about endocranial space and the temporal pole specifically, not a direct measurement of hippocampal volume, which leaves no endocranial impression. This paper examines that anatomical difference in light of two established bodies of evidence: first, that the olfactory system is the only sensory modality with direct, non-thalamic access to the hippocampus and amygdala; and second, that novel odor exposure drives measurable hippocampal and olfactory-bulb neurogenesis in mice (Veyrac et al., 2009). It proposes, as an explicit and falsifiable hypothesis rather than an established conclusion, that *Homo sapiens* possessed a greater neurological capacity for olfactory-driven memory enhancement than Neanderthals. It does not argue that this capacity determined behavior: archaeological evidence indicates Neanderthals used aromatic and medicinal plants despite any anatomical disadvantage (Hardy et al., 2012), and the paper treats culinary and cultural practice as underdetermined by anatomy. The paper closes by naming several unresolved points rather than passing over them: whether fossil olfactory bulb volume reflects heritable evolutionary change or developmental plasticity; whether the rodent neurogenesis mechanism proposed here operates at meaningful levels in the adult human brain, a question itself unresolved in human neuroscience; and how this hypothesis relates to non-anatomical explanations of Neanderthal replacement.

**Keywords:** olfactory bulb; hippocampus; neuroplasticity; *Homo neanderthalensis*; comparative neuroanatomy; cognitive evolution; Deep Symbolic Systems Model

## 1. Introduction

For every sensory modality except olfaction, incoming information is relayed through the thalamus before it reaches cortical and limbic structures. Smell is the exception: olfactory signals pass from the olfactory bulb to the piriform cortex, entorhinal cortex, and amygdala directly, and reach the hippocampus in as few as two synapses, without an obligatory thalamic relay (Chen & Kostka, 2024). This anatomical peculiarity places olfaction in a privileged position relative to memory and emotion, and it raises a specific comparative question: did the two most recent large-brained hominin lineages, *Homo sapiens* and *Homo neanderthalensis*, differ in the neural hardware available to exploit that privileged position?

The present paper argues that they did, on the basis of an existing, peer-reviewed anatomical finding, and proposes a mechanistic hypothesis — not yet tested directly on fossil material — connecting that anatomical difference to differential capacity

for olfactory-driven hippocampal plasticity. This paper is deliberately narrower than an argument about behavior. It does not claim that cooking practices, herb use, or culinary innovation were dictated by brain anatomy in either species; the archaeological record shows clear evidence of Neanderthal use of aromatic and medicinal plants (Hardy et al., 2012), which is not compatible with any simple story in which anatomy alone determines behavior. What is argued here is narrower, and more defensible: that the two species differed in olfactory-related neural capacity, and that this capacity is one plausible, testable factor among the many that shaped the cognitive trajectories of each lineage. This paper extends an argument previously made about the centrality of food acquisition, fire, and the hearth to hominin cognitive and social development (Vondoom, 2026) into a specific sensory-neuroanatomical register.

## **2. Comparative Neuroanatomy: Olfactory Bulbs and the Temporal Pole**

Bastir et al. (2011) used three-dimensional geometric morphometric analysis of the endocranial base — the skeletal surface against which the frontal and temporal lobes and olfactory bulbs develop — to compare *Homo sapiens* and Neanderthal cranial samples. Encephalization occurred independently in both lineages, producing broadly comparable overall brain volumes. Despite this similarity in overall size, the study found that the structures receiving olfactory input were approximately 12% larger in *Homo sapiens* than in Neanderthals, and that *Homo sapiens* also showed a relatively wider orbitofrontal cortex and larger, more forward-projecting temporal lobe poles — a region associated with memory, language, and social cognition (Bastir et al., 2011).

An independent computational reconstruction using a different method — deformation-based modeling of CT-scanned Neanderthal and early *Homo sapiens* skulls — corroborated part of this picture from another angle: Neanderthals showed smaller olfactory bulbs and a less forward-projecting temporal pole alongside a larger occipital region, though that study's central argument concerned cerebellar rather than olfactory differences specifically (Kochiyama et al., 2018). The convergence of two independent methods on smaller Neanderthal olfactory bulbs strengthens the anatomical claim; it does not, on its own, resolve the interpretive question of what that anatomical difference means, which Section 7 addresses directly.

Two precision points matter here. First, 'olfactory bulb' in this literature means the volume of the olfactory bulb fossae on the endocranial cast — a proxy for bulb size, not a direct count of neurons or receptors. Second, the temporal lobe finding concerns the temporal pole specifically, a region implicated in semantic and social cognition, not the hippocampus itself, which sits deeper in the medial temporal lobe and leaves no endocranial impression. No study cited in this paper measured hippocampal volume directly in fossil material, and none could; that should not be inferred from the temporal-pole finding.

Both anchoring studies rest on small fossil samples, and the original authors say so themselves. Bastir et al.'s (2011) full comparative sample included five Neanderthal specimens (La Ferrassie 1, La Chapelle-aux-Saints 1, Saccopastore 1, Guattari 1, Forbes' Quarry) and four fossil *Homo sapiens* specimens (Singa, Skhul V, Mladeč 1, Cioclovina), alongside a larger comparative sample of 75 recent adult humans and 30 chimpanzees used to establish reference ranges. The specific cribriform-plate measurement underlying the widely cited 12% figure was narrower still: it compared three Neanderthals (Forbes' Quarry, Saccopastore 1, Guattari 1; mean 19.16 mm) against a *Homo sapiens* mean of 22.32 mm, using non-parametric statistics because the sample was too small for standard parametric tests. The authors state the limitation directly, in their own words: caution is warranted because of the small Neanderthal sample (Bastir et al., 2011). Kochiyama et al. (2018) used a comparably small sample of four Neanderthal and four early *Homo sapiens* skulls. Point estimates such as the 12% figure describe measurements on a handful of individuals, not population-level statistics with narrow confidence intervals, and both fossil records span considerable time depth and geography that neither study partitions by region or period. With those caveats attached, what the two studies together establish is a structural asymmetry between the species in exactly the two regions most relevant to olfactory-driven memory: the olfactory bulb, and the temporal-pole structures its output eventually reaches.

### **3. The Olfactory Pathway: A Direct Route to the Hippocampus and Amygdala**

Unlike vision, audition, touch, and taste, olfactory information is not relayed through the thalamus before reaching cortex. Olfactory receptor neurons project to the olfactory bulb, which in turn projects directly to the piriform cortex, anterior olfactory nucleus, olfactory tubercle, and the cortical amygdala; piriform and entorhinal cortex then project onward to the hippocampus and prefrontal cortex (Chen & Kostka, 2024). This gives smell privileged, low-synapse access to two structures central to memory formation and emotional salience. It is the physiological basis for the well-documented phenomenon — sometimes called the Proust phenomenon — in which odors trigger unusually vivid, emotionally charged autobiographical memories (Herz & Schooler, 2002), and it is the reason the size and integrity of the olfactory bulb and its downstream temporal targets are directly relevant to a hypothesis about memory capacity, rather than a loose metaphorical connection.

Part of what makes this pathway biologically significant is that some of what it carries does not need to be learned. Hussain et al. (2013) identified a dedicated, high-affinity olfactory receptor for cadaverine, one of two diamine compounds (with putrescine) produced by bacterial decay of animal tissue, and showed that avoidance of this odor is innate rather than learned, conserved across vertebrate species from fish to mammals. This establishes, with direct experimental support, that at least some olfactory signals are treated by the nervous system as inherently and immediately salient — evolutionarily pre-wired survival information rather than information requiring cultural or individual learning to interpret. It does not

establish a specific reaction-time figure in humans, and no such figure is claimed here.

#### **4. The Physics and Physiology of Odor Signal Strength**

Before an odorant can be perceived it must first become airborne. The rate at which a substance evaporates is governed by its vapor pressure, which falls as temperature falls; consequently, the concentration of odorant reaching the nose is directly affected by ambient temperature, and separately by humidity, which affects the dilution and dwell time of volatile compounds in air (Pannunzi & Nowotny, 2019). Independently of this atmospheric-chemistry effect, human olfactory sensitivity itself varies with environmental conditions: a controlled climate-chamber study of 75 volunteers found that odor detection thresholds were lower — sensitivity was higher — in humid conditions than in dry conditions, and were also affected by barometric pressure (Kuehn et al., 2008).

That study measured humidity and pressure directly; it did not test temperature as an independent variable. The inference that cold climates reduce olfactory signal strength — because cold air typically carries less absolute moisture, and because odorant volatility itself falls with temperature — is drawn here from the combination of these two findings rather than from a single study that tested temperature directly. This is flagged explicitly rather than presented as demonstrated.

#### **5. Olfactory Novelty and Hippocampal Neuroplasticity**

A substantial experimental literature in mice shows that olfactory stimulation drives adult neurogenesis in both the olfactory bulb and the hippocampus. Veyrac et al. (2009) found that repeated daily exposure to novel odorants — but not exposure to a stable, enriched olfactory environment — improved short-term olfactory memory and increased neurogenesis in the mouse olfactory bulb, and that this effect depended on noradrenergic signaling. This finding matters for the argument made here in a specific way: it indicates that novelty, not the mere presence of aroma, is the active ingredient — a finding consistent with the idea that a wider range of encountered odorants, rather than any single aromatic substance, would be expected to drive greater plasticity. A related study exposed mice to novel pleasant odors for three weeks and found this increased neuronal cell counts in the hippocampus directly, alongside cell proliferation and neurogenesis in the olfactory bulb, using stereological cell-counting methods (Rusznák et al., 2018).

These findings establish a real, replicated, mechanistic pathway from olfactory novelty to hippocampal and olfactory-bulb plasticity within an individual's lifetime. They do not, on their own, establish that this mechanism produced heritable, evolutionary change in bulb or temporal-lobe size across hominin generations; that inferential gap is addressed directly in Section 7.

A further caveat is worth stating plainly rather than assumed away. Whether adult humans undergo meaningful hippocampal neurogenesis at all is itself an

unresolved, actively contested question in human neuroscience, independent of anything specific to olfaction. Sorrells et al. (2018) reported that new neurons in the human dentate gyrus become undetectable by adulthood; Boldrini et al. (2018), using different tissue-preparation methods, reported that neurogenesis persists throughout aging. This dispute is not resolved at the time of writing. The mechanism proposed in this paper — novelty-driven hippocampal neurogenesis — has strong, repeatedly replicated support in mice. Its relevance to adult humans specifically, which an evolutionary argument about *Homo sapiens* ultimately requires, cannot be assumed on the basis of the rodent literature; it is an open empirical question this hypothesis inherits rather than resolves.

## **6. Synthesis: A Capacity Hypothesis, Not a Behavioral Determinism**

Bringing Sections 2 through 5 together: *Homo sapiens* possessed, on the available fossil evidence, a measurably larger olfactory bulb and larger temporal-pole structures than Neanderthals (Bastir et al., 2011; Kochiyama et al., 2018); the olfactory system has privileged, thalamus-independent access to the hippocampus and amygdala (Chen & Kostka, 2024); and novel-odor exposure is known to drive hippocampal and olfactory-bulb neurogenesis in a living mammalian brain, at least in mice (Veyrac et al., 2009). The hypothesis proposed here is that a larger olfactory bulb and larger downstream temporal-pole targets constituted a greater neural capacity in *Homo sapiens* for converting olfactory novelty into hippocampal plasticity — a difference in instrument, not a difference in program.

One bridging step in this synthesis is not itself demonstrated in the cited literature and should be named rather than left implicit: the move from measured anatomical size to a claim about plasticity capacity. Veyrac et al. (2009) shows that novel exposure drives plasticity; it does not show that a larger baseline bulb predicts more plasticity per unit of novel exposure. The inference from 'larger structure' to 'greater capacity for structure-dependent plasticity' is a reasonable working assumption grounded in general principles of neural tissue — more processing units, more possible connections — but it is an assumption this paper is making, not a specific finding it can cite. One partial, though incomplete, support exists: in living humans, olfactory bulb volume correlates positively with measured odor threshold, discrimination, and identification performance (Buschhüter et al., 2008). That is a within-species, within-lifetime structure-function correlation, not a cross-species evolutionary one, and it says nothing about plasticity capacity specifically — but it does support treating bulb size as functionally meaningful rather than a size difference with no known behavioral correlate.

This is deliberately framed as a capacity claim rather than a behavioral one. Archaeological evidence shows Neanderthals used aromatic plants, including chamomile and yarrow, apparently for flavor or medicinal purposes, and processed a range of plant foods thermally (Hardy et al., 2012). Whatever anatomical difference existed did not prevent Neanderthal engagement with aromatic material; it may instead have set a ceiling on how much cognitive benefit that engagement could produce, or on how strong the reward signal driving further exploration was.

What either species actually did with the sensory information available to them is a matter of behavior, culture, and individual and group variation — properties this paper does not claim anatomy determines.

## **7. Limitations and Open Questions**

Five limitations deserve explicit statement here, in keeping with this paper's standard of flagging rather than silently passing over unresolved points. It is also worth situating this kind of claim historically before turning to specifics: endocast-based inference from skull shape to cognitive capacity has a long and uneven track record, including characterizations from the 1800s and early 1900s of Neanderthals as cognitively brutish that later evidence substantially revised. This paper's claim is narrower and more qualified than those earlier arguments, but it belongs to the same general category of evidence, and that history is a reason for caution rather than a reason to avoid the question.

First, and most importantly: a published comment on the original Bastir et al. (2011) article noted that clinical MRI literature — cited by the commenter in relation to conditions such as laryngectomy and chronic rhinosinusitis — has linked olfactory bulb volume to environmental and pathological factors during an individual's lifetime, not only to genetically fixed, evolved structure, and argued that this casts some doubt on a purely evolutionary reading of fossil endocranial bulb-cavity size. I have not independently verified the specific clinical studies referenced in that comment, and I have not located a subsequent peer-reviewed rebuttal addressing it. It is noted here as an open methodological concern rather than a settled critique, and it means the 12% figure should be read as a measured difference in endocranial cavity space historically occupied by the olfactory bulb, not as a proven, purely genetic difference in adult olfactory capacity between the two species.

Second, this paper has not established — and does not claim to have established — that individual-lifetime neuroplasticity of the kind demonstrated by Veyrac et al. (2009) in mice translates into heritable, trans-generational anatomical change of the kind measured by Bastir et al. (2011) in the fossil record. Connecting a developmental plasticity mechanism to an evolutionary outcome requires a selection argument — heritability of bulb size, a demonstrated fitness benefit tied to olfactory memory capacity, and sufficient generational depth for selection to act — that this paper has not supplied.

Third, and most consequential for how much weight this paper's central claim can bear: a 2026 reanalysis directly addressed how much cognitive difference endocranial shape differences between the species should be expected to predict, and concluded that they predict a cognitive effect size of 0.14 standard deviations or less — smaller than the variation observed among different modern human populations, which is not generally treated as evolutionarily or cognitively significant (Schoenemann et al., 2026). That analysis was not conducted on the olfactory bulb and temporal-lobe measurements specifically; it addressed endocranial shape differences more broadly. But it is directly relevant here, because

it argues that endocast-based anatomical differences between the two species, as a category of evidence, may simply not carry the behavioral or cognitive significance that has often been attributed to them. This paper's hypothesis should be read with that caution attached: the anatomical difference at its center is real and measured, but its functional consequences — for either species, in either direction — remain genuinely open.

A falsifiable prediction that would help close this gap: if larger olfactory bulbs were under active selection in the sapiens lineage specifically, bulb-cavity size should show a directional trend across the sapiens fossil record correlated with time depth, rather than appearing uniform from the earliest anatomically modern specimens onward. To my knowledge this specific trend has not yet been tested in the published literature, and it is offered here as a research direction rather than a claim. Testing it with current fossil sample sizes — a handful of specimens per study, as noted above — would likely be underpowered to detect anything short of a large effect; a more immediately tractable test might instead compare olfactory bulb fossae volume against isotopic or archaeological proxies for dietary and environmental aromatic exposure within the existing sapiens fossil sample, rather than waiting on new fossil discoveries.

Fourth, this paper has not situated its hypothesis relative to non-anatomical explanations of Neanderthal replacement, and should. A 2025 mathematical model shows that recurrent small-scale interbreeding between demographically larger *Homo sapiens* populations and smaller Neanderthal populations could account for Neanderthals' genetic absorption within 10,000–30,000 years through neutral genetic drift alone, without requiring any cognitive, behavioral, or selective advantage on either side (Amadei et al., 2025). This model does not compete with the hypothesis proposed here in a strict sense — a genetic-dilution account of population replacement and an anatomical-capacity account of individual cognitive difference are not mutually exclusive — but it is a reminder that this paper's hypothesis is one candidate factor among several current, actively studied explanations, not a freestanding account of why Neanderthals disappeared. Nothing in this paper should be read as claiming otherwise.

Fifth, and worth stating with particular care: the authors of the anchoring anatomical study explicitly warned against the interpretive move this paper makes. Bastir et al. (2011) caution that their olfactory bulb and cribriform plate findings should not be read as evidence of a better capacity of smell in *Homo sapiens* relative to Neanderthals, of the kind seen between macrosomatic and microsomatic mammal species; they favor a more conservative reading in which olfactory-related circuitry was favored as part of a broader, integrated neural network serving social cognition, not smell capacity as such. This paper's central hypothesis, framed explicitly around capacity for olfactory-driven memory enhancement, goes further than the original anatomical study itself is willing to go. That does not make the hypothesis wrong — the neuroplasticity mechanism in Section 5 and the memory-specific pathway in Section 3 give the capacity framing here a different evidentiary basis than a simple smell-acuity comparison would. But readers should know this paper extends past where its own anchoring source stops, and should weigh the hypothesis accordingly.

## 8. Relation to the Deep Symbolic Systems Model and Prior Work

This paper extends an argument developed in Vondoom (2026), which situated cooking, fire, and the hearth as an underexamined convergence point for cognitive, social, and economic drivers of civilizational complexity, predating monuments and writing by hundreds of thousands of years. That paper treated food acquisition and preparation as a general driver of hominin cognitive architecture; the present paper narrows the focus to a single sensory channel — olfaction — and a single proposed mechanism — hippocampal neuroplasticity — as one candidate contributor among the many the earlier paper surveys.

Within the Deep Symbolic Systems Model (DSSM; Vondoom, 2027), embodied practices stabilize recursively into durable symbolic systems across successive developmental stages. A greater capacity for olfactory-driven memory consolidation, of the kind proposed here, would sit at the earliest, pre-symbolic stage of that framework: it is a proposed biological precondition for the kind of repeated, memory-dependent behavioral stabilization — the finding, marking, and returning to a food source; the retention and reproduction of a preparation technique — that DSSM treats as the raw material that later solidifies into symbolic form.

One clarification of scope: DSSM, as documented in Vondoom (2027), is the author's own peer-reviewed theoretical framework. It is invoked in this section as context for where this paper's hypothesis fits within a broader research program — not as independent empirical support for the anatomical or mechanistic claims argued earlier in the paper, which rest on the sources cited in Sections 2 through 7.

## 9. Sustainable Development Goals Relevance

**SDG 3 — Good Health and Well-Being:** the neuroplasticity mechanism discussed in Section 5 (novelty-driven olfactory enrichment and hippocampal neurogenesis) is directly relevant to ongoing clinical research into olfactory training as a low-cost intervention for age-related cognitive decline and post-viral anosmia.

**SDG 4 — Quality Education:** by tracing memory capacity to a specific, testable neuroanatomical mechanism rather than treating cognitive evolution as an unexplained given, this paper models falsifiable, evidence-graded reasoning about deep human history relevant to science education and public science communication.

**SDG 15 — Life on Land:** comparative hominin neuroanatomy depends on continued paleoanthropological fieldwork and the preservation of fossil sites and associated environmental records, connecting this research to the broader conservation of terrestrial paleoecological heritage.

## 10. Conclusion

The comparative anatomical record shows a real, measured difference between *Homo sapiens* and Neanderthals in exactly the neural structures most relevant to olfactory-driven memory. The mechanistic literature on olfactory neuroplasticity shows a real, replicated pathway connecting odor novelty to hippocampal growth in living mammals. Connecting the two — proposing that this anatomical difference gave *Homo sapiens* a greater capacity, not a behavioral guarantee, for turning olfactory experience into memory — is this paper's contribution, offered explicitly as a hypothesis with named limitations and a stated path toward falsification, not as a settled account of why one lineage persisted and the other did not.

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## AI Disclosure

This paper was developed with the assistance of Claude (Anthropic), an artificial intelligence language model, used as a drafting and literature-verification assistant. The intellectual framework, core arguments, and scholarly interpretation are those of the author. All citations have been checked against primary or indexed secondary sources; points of uncertainty and unresolved tension in the underlying literature are flagged explicitly throughout — including within Sections 2, 4, 5, and 6 — and consolidated in Section 7, rather than resolved silently. This disclosure is made in accordance with the author's standard practice of transparent AI use across all published work.

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